

Computer Vision – Lecture 4

Image Processing III

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Course Outline

- Image Processing Basics
 - Image Formation
 - Linear Filters
 - Gradient Filters and [Edge Detection](#)
- Segmentation
- Object Recognition and Categorization
- Deep Learning Approaches for Vision
- 3D Reconstruction

Topics of This Lecture

- Edge detection
 - Image gradients
 - Canny edge detector



Edge Detection

- Goal: map image from 2D array of pixels to a set of curves or line segments or contours.
- Why?

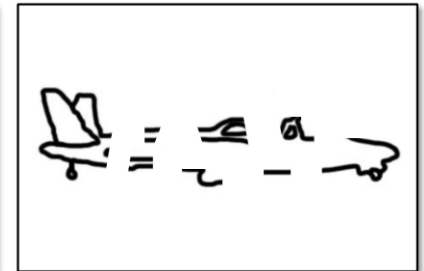
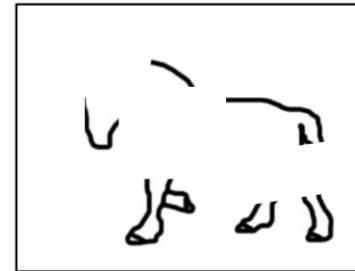
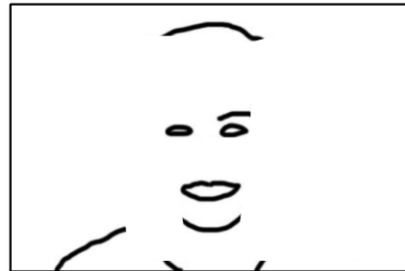


Figure from J. Shotton et al., PAMI 2007

- Main idea: look for strong gradients, post-process

Designing an Edge Detector

- Criteria for an “optimal” edge detector:
 - **Good detection**: the optimal detector should minimize the probability of false positives (detecting spurious edges caused by noise), as well as that of false negatives (missing real edges).
 - **Good localization**: the edges detected should be as close as possible to the true edges.
 - **Single response**: the detector should return one point only for each true edge point; that is, minimize the number of local maxima around the true edge.



True
edge



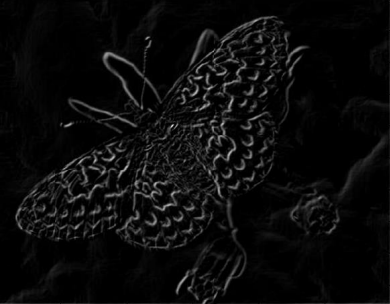
Poor robustness
to noise



Poor
localization



Too many
responses



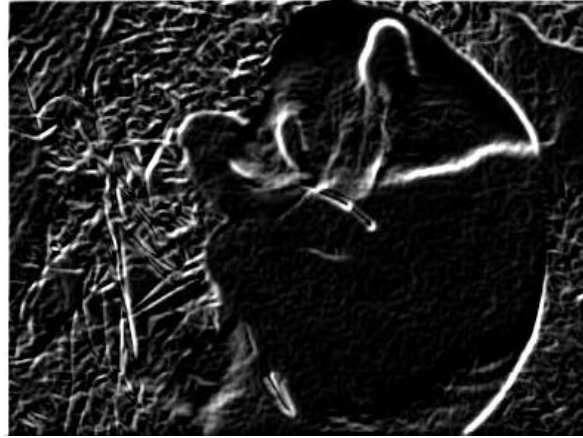
Gradients $\rightarrow$ Edges



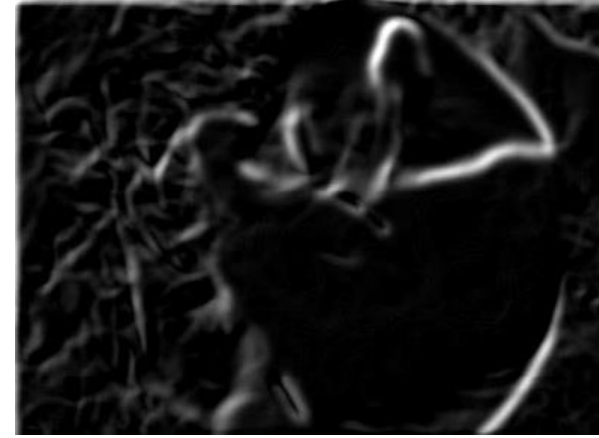
Primary edge detection steps

1. Smoothing: suppress noise
 2. Edge enhancement: filter for contrast
 3. Edge localization
 - Determine which local maxima from filter output are actually edges vs. noise
 - Thresholding, thinning
- Two issues
 - At what scale do we want to extract structures?
 - How sensitive should the edge extractor be?

Scale: Effect of σ on Derivatives



$\sigma = 1$ pixel



$\sigma = 3$ pixels

- The apparent structures differ depending on Gaussian's scale parameter.

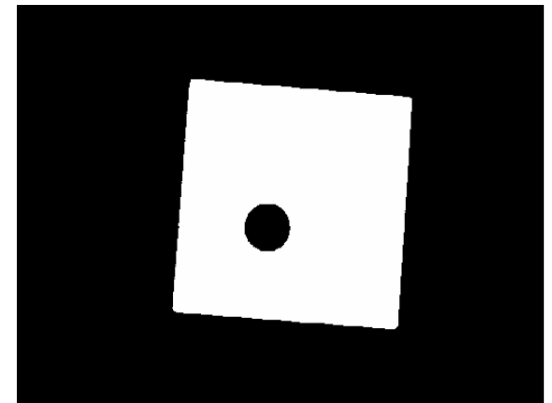
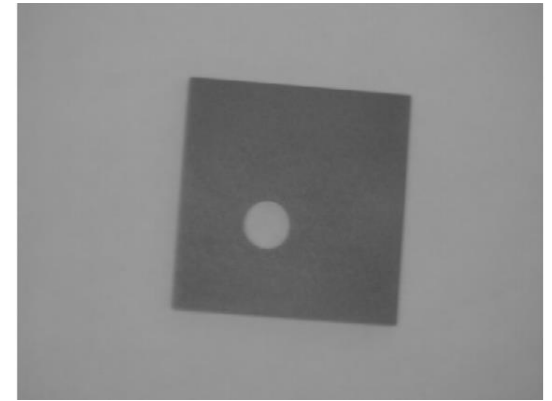
⇒ Larger values: larger-scale edges detected

⇒ Smaller values: finer features detected

Sensitivity: Compare to Thresholding

- Choose a threshold t
- Set any pixels less than t to zero (off).
- Set any pixels greater than or equal t to one (on).

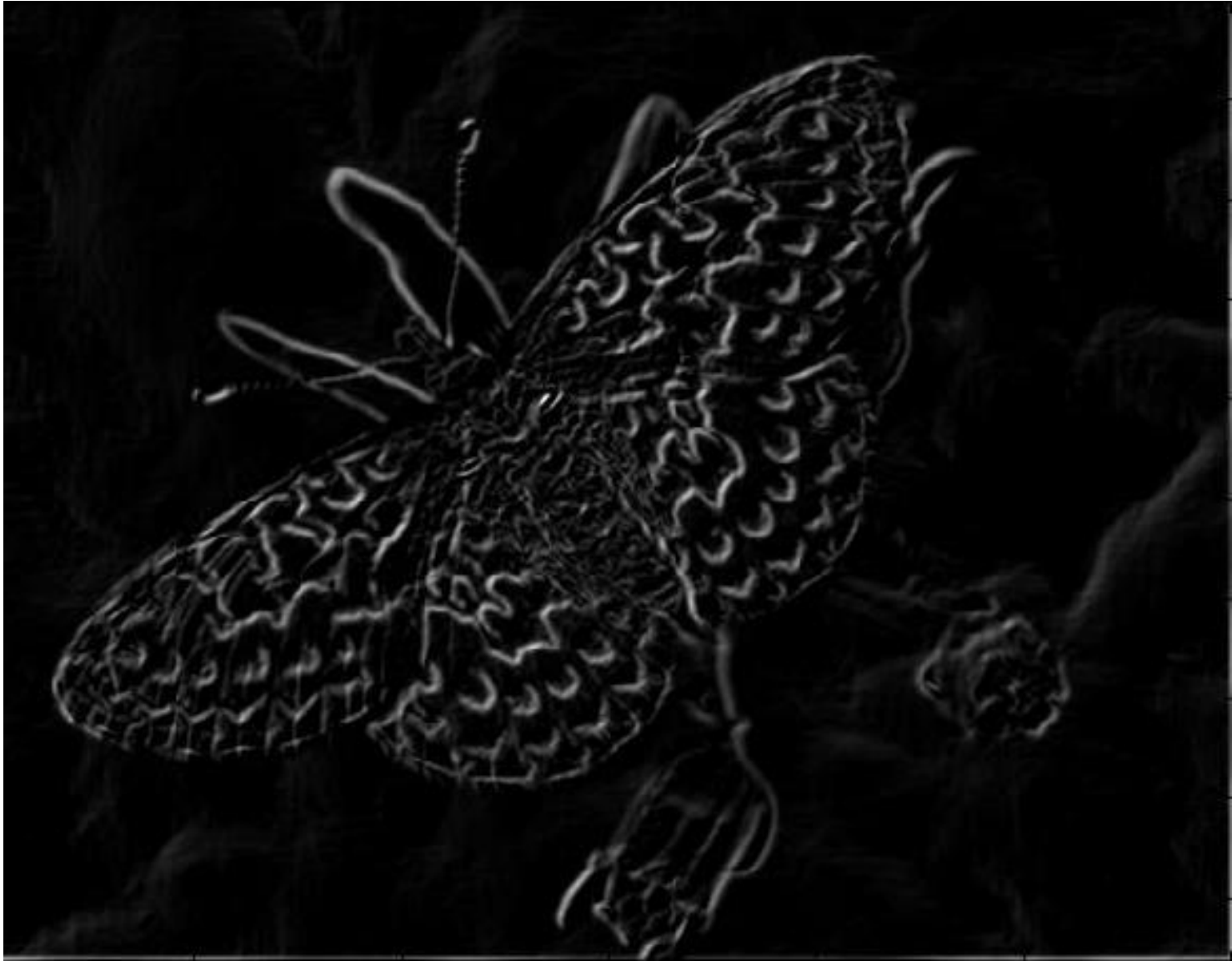
$$F_T[i, j] = \begin{cases} 1, & \text{if } F[i, j] \geq t \\ 0, & \text{otherwise} \end{cases}$$



Original Image



Gradient Magnitude Image



Slide credit: Kristen Grauman

Thresholding with a Lower Threshold



Thresholding with a Higher Threshold



Canny Edge Detector

- A very widely used edge detector in computer vision
- Theoretical model: step-edges corrupted by additive Gaussian noise
- Canny has shown that the first derivative of the Gaussian closely approximates the operator that optimizes the product of *signal-to-noise ratio* and localization.

J. Canny, [A Computational Approach To Edge Detection](#), *IEEE Trans. Pattern Analysis and Machine Intelligence*, 8:679-714, 1986.

The Canny Edge Detector



Original image

The Canny Edge Detector



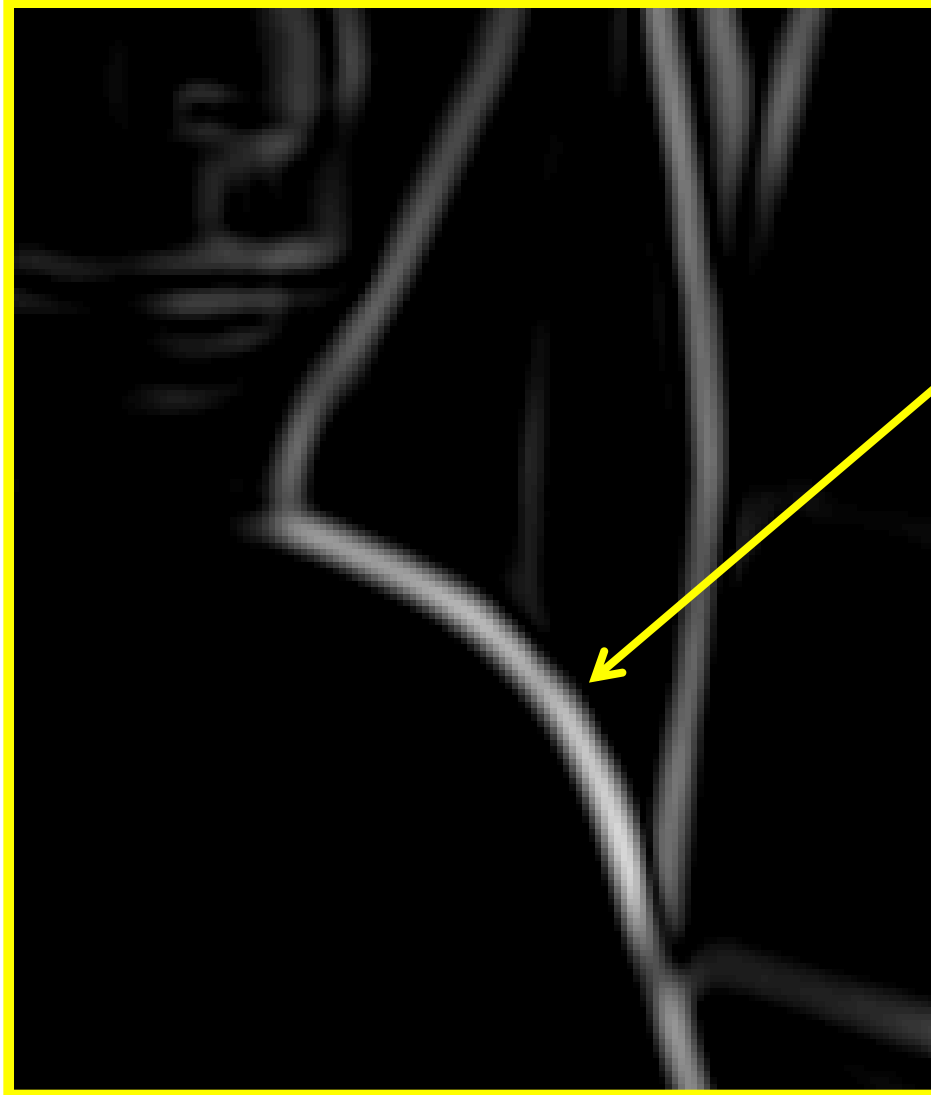
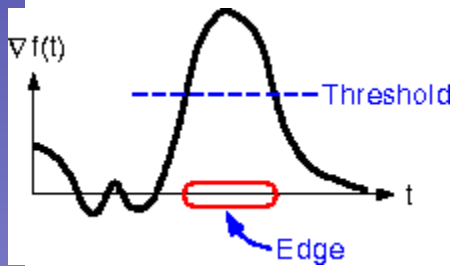
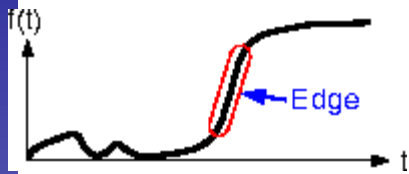
Gradient magnitude

The Canny Edge Detector



Thresholding

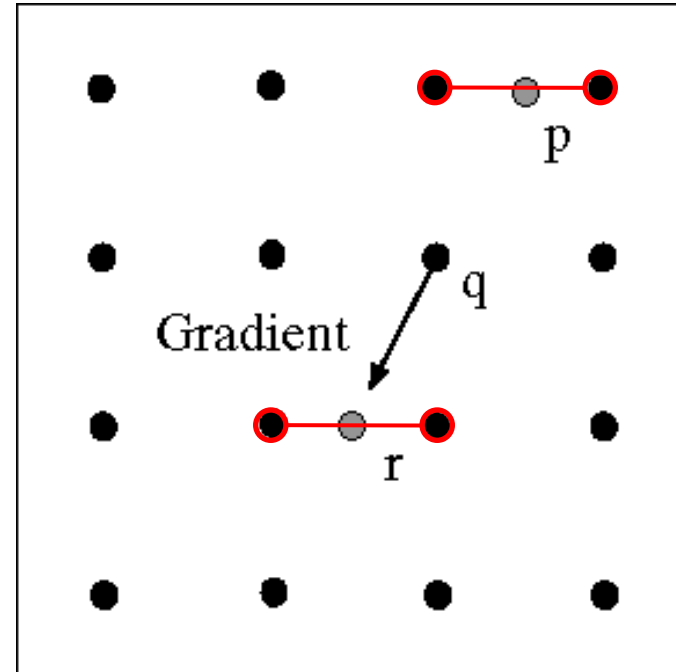
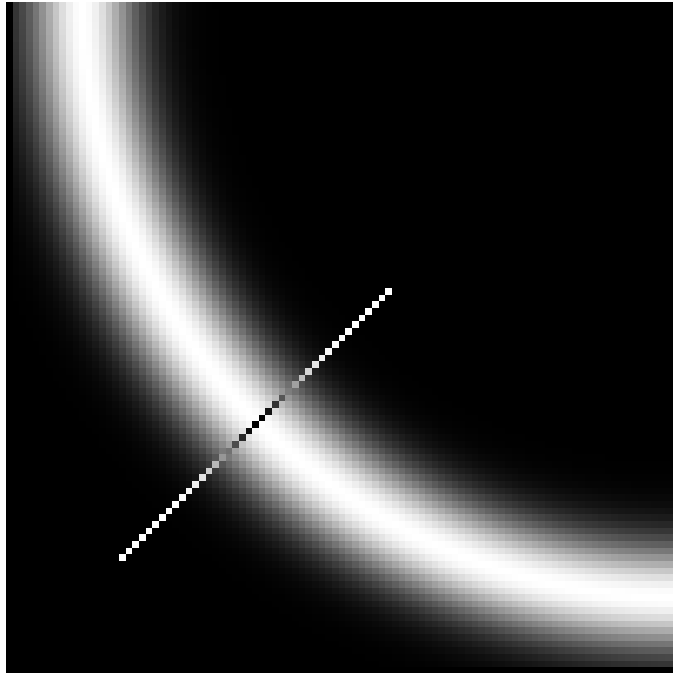
The Canny Edge Detector



How to turn these thick regions of the gradient into single-pixel curves?



Non-Maximum Suppression



- Check if pixel is local maximum along gradient direction, select single max across width of the edge
 - I.e., keep q iff $\text{Mag}(q) > \text{Mag}(p)$ and $\text{Mag}(q) > \text{Mag}(r)$.
 - Requires checking interpolated pixels p and r $\Rightarrow$ Linear interpolation based on gradient direction

The Canny Edge Detector



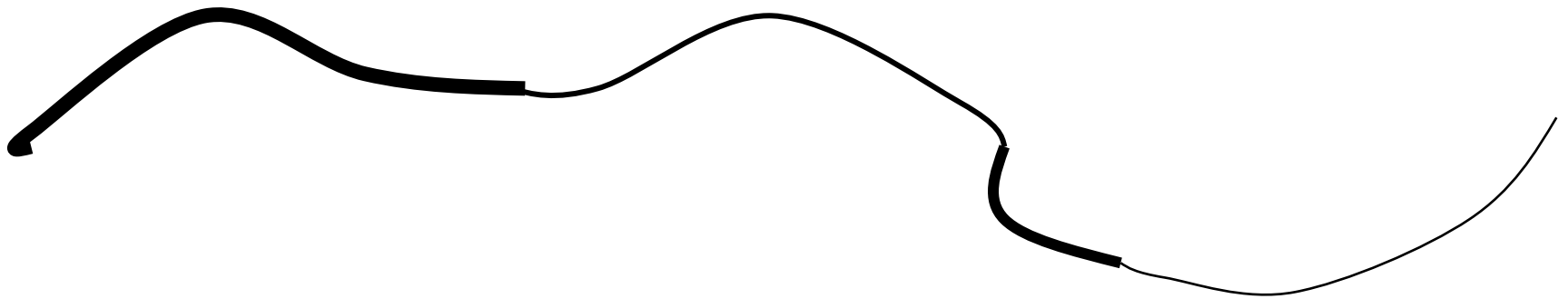
Problem: pixels along this edge didn't survive the thresholding.

Thinning
(non-maximum suppression)

Solution: Hysteresis Thresholding

- Hysteresis: A lag or momentum factor
- Idea: Maintain two thresholds k_{high} and k_{low}
 - Use k_{high} to find strong edges to start edge chain
 - Use k_{low} to find weak edges which continue edge chain
- Typical ratio of thresholds is roughly

$$k_{high} / k_{low} = 2$$



Hysteresis Thresholding



Original image



High threshold
(strong edges)



Low threshold
(weak edges)



courtesy of G. Loy

Hysteresis threshold

Summary: Canny Edge Detector

1. Filter image with derivative of Gaussian
2. Find magnitude and orientation of gradient
3. Non-maximum suppression:
 - Thin multi-pixel wide “ridges” down to single pixel width
4. Linking and thresholding (hysteresis):
 - Define two thresholds: low and high
 - Use the high threshold to start edge curves and the low threshold to continue them

Object Boundaries vs. Edges



Background

Texture

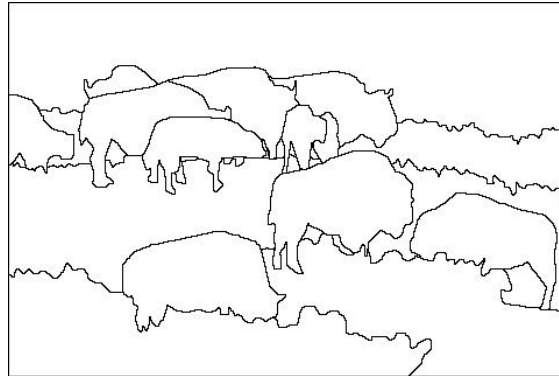
Shadows

Edge Detection is Just the Beginning...

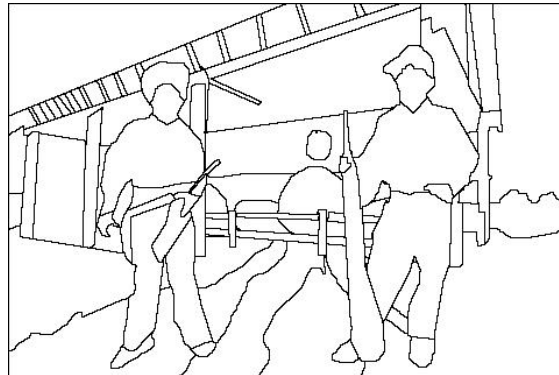
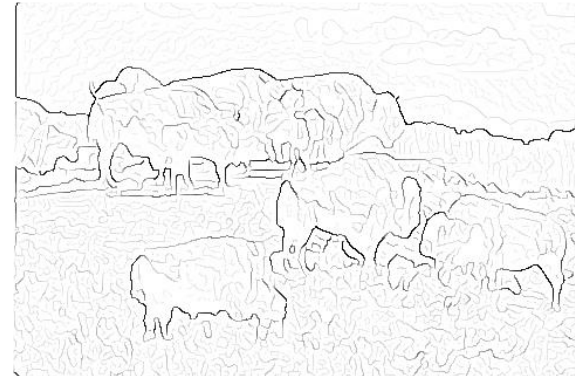
Image



Human segmentation



Gradient magnitude



- Berkeley segmentation database:
<http://www.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/segbench/>

References and Further Reading

- Background information on image gradients is available in Chapter 3; background on edge detection can be found in Chapter 4 of the Szeliski book.

R. Szeliski
Computer Vision – Algorithms and Applications
Springer, 2010
([draft version available online](#))

